

Julian Coda-Forno

Mountain View – USA

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Languages: French (native) | Spanish (native) | English (fluent)

Research Interests

Large Language Models; Meta-Learning; Cognitive Science; Deep Learning; In-context Learning; Reinforcement-Learning; LLM Reasoning; Decision-making; Mechanistic Interpretability

Education

ELLIS & Technical University of Munich (TUM)

Munich & London

PhD in Machine learning

2022–Present

Supervisors: Dr. Eric Schulz (Helmholtz Munich) & Dr. Jane X. Wang (Google DeepMind)

Topic: LLMs' representations, behaviour and decision-making from a cognitive science perspective

University College London (UCL)

London, UK

MS.c in Data Science & Machine Learning, Distinction (82.6%) with Dean's List

2020–2021

Supervisors: Prof. Neil Burgess (UCL) & Dr. Zafeirios Fountas (Huawei)

Title: "Leveraging episodic memory in model-based RL"

University of Manchester

Manchester, UK

BEng Aerospace Engineering, First class Honours

2015–2019

Experience

Google Deepmind

Mountain View, USA

Student Researcher

2025–Present

Meta Platforms, Inc

Menlo Park, USA

Research Scientist Intern (6 months)

2025

Working on reasoning in LLMs

Huawei

London, UK

Visiting Research intern (6 months)

2021–2022

Collaborated with Huawei Neuromorphic Computing Group for my UCL Master's research thesis and pursued the research further. Investigated how the neuroscience concept of episodic memory could be used in model-based reinforcement learning for more sample efficiency in complex tasks and environments.

Accepted paper at MemARI (2022 NeurIPS workshop)

Rolls-Royce Plc

Birmingham, UK

Software Engineering Intern (12 months)

2018–2019

Supervision

Natalia Scharfenberg: Osnabrück University, Master Thesis: "LLM's representations for RL" 2023

Awards

Dean's List: top 5% of student achievement within the faculty of Engineering 2021
Distinctions of the jury: 14th French Olympiads of mathematics - Nice Academy 2014

Summer schools

MIT Brains, Minds & Machines Summer Course: Woods Hole, USA 2023

Invited talks

Harvard Efficient-ML seminar series
"Rising star speaker" - Tutorial on meta-learning in deep neural networks 2024

Reviewing

Conference on Neural Information Processing Systems (NeurIPS) 2024
Conference on Neural Information Processing Systems (NeurIPS) 2023

Research papers

- [1] **Julian Coda-Forno** Marcel Binz Akshay K. Jagadish, Mirko Thalmann and Eric Schulz. Meta-learning ecological priors from large language models explains human learning and decision making. *arXiv:2509.00116*, 2025.
- [2] Marcel Binz, ..., **Julian Coda-Forno**, ..., and Eric Schulz. A foundation model to predict and capture human cognition. *Nature*, 2025.
- [3] **Julian Coda-Forno**, Marcel Binz, Jane X. Wang, and Eric Schulz. Cogbench: a large language model walks into a psychology lab. *International Conference on Machine Learning (ICML)*, 2024.
- [4] Akshay K. Jagadish, **Julian Coda-Forno**, Mirko Thalmann, Eric Schulz, and Marcel Binz. Ecologically rational meta-learned inference explains human category learning. *International Conference on Machine Learning (ICML)*, 2024.
- [5] **Julian Coda-Forno**, Kristin Witte, Akshay K Jagadish, Marcel Binz, Zeynep Akata, and Eric Schulz. Inducing anxiety in large language models increases exploration and bias. *arXiv:2304.11111*, 2023.
- [6] Elif Akata, Lion Schulz, **Julian Coda-Forno**, Seong Joon Oh, Matthias Bethge, and Eric Schulz. Playing repeated games with large language models. *Nature Human Behaviour*, 2023.
- [7] **Julian Coda-Forno**, Marcel Binz, Zeynep Akata, Matt Botvinick, Jane Wang, and Eric Schulz. Meta-in-context learning in large language models. *Advances in Neural Information Processing Systems*, 36, 2023.
- [8] **Julian Coda-Forno**, Changmin Yu, Qinghai Guo, Zafeirios Fountas, and Neil Burgess. Leveraging episodic memory to improve world models for reinforcement learning. *Memory in Artificial and Real Intelligence (MemARI) workshop at NeurIPS*, 2022.